

$c \rightarrow D^0$ Fragmentation Functions

The code offers three interchangeable models for the charm-quark-to- D^0 fragmentation function $D(z)$, selected by `par->frag_type` (`FragmentationType::BCFY` / `KnieshlKramer` / `LHAPDF`, default `BCFY`). In every case, the cross-section integrand uses the combination

$$\text{weight}(z_h) = \frac{D(z_h)}{z_h^2}, \quad z_h \in [z_{\min}, z_{\max}] = [0.05, 1] \quad (1)$$

`integrand.cpp:31--43`, $z_{\min}, z_{\max}$: `main.cpp:96--97` where z_h is the momentum fraction of the charm quark carried by the D^0 meson ($p_{D^0} = z_h p_c$).

1 BCFY (Braaten–Cheung–Fleming–Yuan) – default

This model splits the fragmenting charm quark into a pseudoscalar piece (D^0 directly) and a vector piece ($D^{*0} \rightarrow D^0$), each following the perturbative-QCD BCFY functional form, built from a single non-perturbative shape parameter r .

Pseudoscalar function $D_P(z, r)$.

$$D_P(z, r) = \frac{r z (1 - z)^2}{[1 - (1 - r)z]^6} P(z, r), \quad 0 < z < 1 \quad (2)$$

with

$$\begin{aligned} P(z, r) = & 6 - 18(1 - 2r)z \\ & + (21 - 74r + 68r^2)z^2 \\ & - 2(1 - r)(6 - 19r + 18r^2)z^3 \\ & + 3(1 - r)^2(1 - 2r + 2r^2)z^4 \end{aligned} \quad (3)$$

`fragmentation.cpp:9--23`

Vector function $D_V(z, r)$.

$$D_V(z, r) = \frac{3 r z (1 - z)^2}{[1 - (1 - r)z]^6} V(z, r), \quad 0 < z < 1 \quad (4)$$

with

$$\begin{aligned} V(z, r) = & 2 - 2(3 - 2r)z \\ & + 3(3 - 2r + 4r^2)z^2 \\ & - 2(1 - r)(4 - r + 2r^2)z^3 \\ & + (1 - r)^2(3 - 2r + 2r^2)z^4 \end{aligned} \quad (5)$$

`fragmentation.cpp:28--41`

Combination into $D_c^{D^0}(z)$. The two channels are combined with branching-fraction-like weights, with the vector piece rescaled in argument by the D^{*0}/D^0 mass ratio (to convert "fraction of D^{*0} momentum" into "fraction of D^0 momentum") and restricted to the kinematic threshold $z \leq m_{D^0}/m_{D^{*0}}$:

$$D_c^{D^0}(z) = 0.168 D_P(z, r) + 0.39 \theta\left(\frac{m_{D^0}}{m_{D^{*0}}} - z\right) \frac{m_{D^{*0}}}{m_{D^0}} D_V\left(\frac{m_{D^{*0}}}{m_{D^0}} z, r\right) \quad (6)$$

fragmentation.cpp:45--63

Parameter	Value
r (non-perturbative shape)	0.1 <code>main.cpp:93</code>
m_{D^0}	1.8648 GeV
$m_{D^{*0}}$	2.0067 GeV
Pseudoscalar weight	0.168
Vector weight	0.39

2 Kniehl–Kramer

A simpler, purely phenomenological two-parameter form:

$$D(z) = N \frac{z(1-z)^2}{[(1-z)^2 + \epsilon z]^2}, \quad 0 < z < 1 \quad (7)$$

fragmentation.cpp:66--72

Parameter	Value
N	0.694 <code>main.cpp:94</code>
ϵ	0.101 <code>main.cpp:95</code>

3 LHAPDF

Rather than an analytic form, $D(z)$ is read from an external LHAPDF fragmentation-function grid (member file, default `data/prompt-D0-1-109/prompt-D0-1-109_0000.dat`) for the charm-quark flavor (PDG id 4), interpolated in z at a fixed factorization scale

$$Q = \kappa \sqrt{p_{D^0}^2 + m_c^2}, \quad \kappa = \text{SCALE_FACTOR (default 1)} \quad (8)$$

with a floor $Q \geq m_c$ (below the charm-production threshold the grid is undefined, so the scale is not allowed to drop under m_c even for $\kappa = 0.5$ at very low p_{D^0}). $\kappa = 0.5, 2$ give the conventional scale-variation envelope around the central ($\kappa = 1$) result; the 101 members of the grid (one central + 100 replicas) give the PDF-style replica uncertainty band when swapped in one at a time.

`main.cpp:107--128`